

Technical Document #92: Deep Learning - Comprehensive Overview

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Attention mechanisms allow models to focus on different parts of the input when producing output. They have become essential in sequence-to-sequence models.

Residual connections enable training of very deep neural networks by providing shortcuts for gradient flow. They were introduced in the ResNet architecture.

Long short-term memory (LSTM) networks are a type of RNN that can learn long-term dependencies. They use gating mechanisms to control the flow of information.

Recurrent neural networks (RNNs) are designed to process sequential data. They maintain hidden states that capture information about previous elements in the sequence.

Augmented reality overlays digital information on the physical world. It has applications in training, maintenance, and entertainment.

The Internet of Things (IoT) connects billions of devices worldwide. This connectivity generates massive amounts of data that can be analyzed for insights.

Remote work technologies have transformed the workplace. Collaboration tools and cloud services enable distributed teams to work effectively from anywhere.

Key Takeaways:

- Attention mechanisms allow models to focus on different parts of the input when producing output.
- Convolutional neural networks (CNNs) are designed to process grid-like data such as images.
- Recurrent neural networks (RNNs) are designed to process sequential data.